

SLEEP MATTERS

Two exhausted boys grab some rest between Kung Fu exercises at the Shaolin Tagou Academy, China's largest martial arts school.



GOALS

IN THIS UNIT, YOU WILL:

- Read about the problems of not getting enough sleep.
- Learn about one person's experience of sleep deprivation.
- Explore current research on sleep.

THINK AND DISCUSS

1. Why do you think we need to sleep?
2. Do you think certain people need more sleep than others? If so, why?

Lesson A

PRE-READING

A. Note answers to the questions below, and share your ideas with a partner.

1. What factors do you think help you get a good night's sleep?

Answers will vary. Sleep is aided by eating well, staying stress-free, not overstimulating yourself at night, etc.

2. How many hours of sleep do you think people your age typically get each night?

Answers will vary.

3. Look at the infographic on page 28. According to the information given, how much sleep should you get every night? Do you get this much sleep? If not, why not?

Answers will vary depending on age. Adults, age 18 or over, need seven to nine hours per night.

- B. Look at the two headings in the reading. Predict the information that each section might contain.**

Predictions will vary. Actual answers are:

1. Why do many people feel they don't have enough time to sleep?

We live in a 24/7 culture now where things are available and places are open all the time. Also, people work long hours, which cuts into their sleep time.

2. What are some effects of lack of sleep, or "sleep deprivation"?

The effects of no sleep include poor performance at work and school. Fatigue is also a major cause of dangerous accidents, such as car crashes.

A tired student takes a nap on a classroom desk during lunch break. Lack of sleep is a common problem among young adults.



Are You Sleeping Enough?

We've all had restless nights when we couldn't get to sleep—and have suffered the next day as a result. Sleep experts say the average adult requires seven to nine hours of sleep per night; in fact, anything less could harm our health.



NO TIME TO SLEEP

In spite of the importance of sleep, many people are not getting enough. For example, Americans sleep one hour less per night on average than they did 20 to 30 years ago. “The dependency on caffeine and the whole Starbucks culture is certainly one proof that our society is sleepier than ever before,” says William Dement, a sleep researcher at Stanford University. Sleep is a biological need, much like food and water. If totally deprived of shut-eye, humans **ultimately** perish. Yet millions of people are increasingly skimping on their sleep.

Why has sleep deprivation become such a big problem? There are a couple of possible reasons. Researchers say that today’s sleep deprivation is partly due to modern expectations. These days, consumers expect information, entertainment, and food to be available round-the-clock. One of the results of this 24/7 culture is that consumers stay awake longer and later than they used to. Consequently, workers who provide us with goods and services can’t stop working when the sun goes down. In addition, the fast-paced and competitive world of business often rewards employees for arriving at work early and staying late. As a result, many employees are willing to skip sleep in order to impress the boss.



EFFECTS OF SLEEP DEPRIVATION

Sleep deprivation can have a significantly detrimental impact on our performance, however. According to Charles Czeisler of Harvard University, staying awake for 24 hours or sleeping only five hours a night for a week is like having a blood-alcohol level of 0.1 percent. That is well above the **legal** limit for safe driving in most countries. Most people probably wouldn't get drunk before going to work or taking an exam; but if they are sleeping an average of five hours a night, they may be working, studying, driving, and making decisions as though they are intoxicated.

Sleep deprivation can result in poor grades or poor performance at work. But in some cases, it can have more dangerous effects. In February 2009, a commuter plane crashed on the way from Newark, New Jersey, to Buffalo, New York, killing all 49 people on board and one person on the ground. An **investigation** determined that the crash was most likely a result of fatigue. The copilot, and probably the pilot, had slept very little the day before the crash. Not surprisingly, sleep deprivation is linked to car accidents, too. The U.S. **Institute** of Medicine reports that almost 20 percent of car accidents happen because drivers are sleepy.



Sleep deprivation can be a particularly grave problem in the **medical** profession. In 2007, Czeisler researched the habits of 2,700 first-year medical residents. Most of these residents worked for extremely long periods, some working 30-hour-long shifts twice a week. Many admitted that fatigue had had severe **consequences** on their performance and treatment of patients. Twenty percent of residents **acknowledged** making fatigue-related mistakes that led to **injury** of a patient. Five percent confessed that lack of sleep caused them to make a mistake that resulted in a patient's death.





A man taking a nap on a train car in Cairo, Egypt. According to sleep researcher Carl Hunt, “Sleep is just as important for our overall health as are exercise and a healthy diet.”



We all know that we need sleep, but sleep deprivation continues to be a widespread problem around the world. Studies show that one in five adults suffer from daytime sleepiness. Among those aged 18 to 34, 50 percent say that daytime sleepiness interferes with their daily work. To many, the ability to function without much sleep is something to brag about. However, sleep is **crucial** to our well-being and our performance. It may be time to wake up and change our perspective on sleep.

brag: *n.* to talk about your achievements in a boastful or proud way

detrimental impact: *adj.* damaging or harmful effect

exhaustion: *n.* a state of extreme physical and/or mental tiredness

grave: *adj.* very serious

medical residents: *n.* medical school graduates undergoing on-the-job training



DEADLY DEPRIVATION

A complete lack of sleep can be deadly. In a sleep study performed by the prominent sleep researcher Allan Rechtschaffen at the University of Chicago, rats were kept awake and not permitted to sleep at all. After about two weeks, all of the rats died. The rats' organs were not damaged, and the rats didn't seem to have anything seriously wrong with them. **Apparently**, they all died from exhaustion.

Developing Reading Skills

GETTING THE MAIN IDEAS

Read the first sentence of a summary of the reading passage. Complete the summary by choosing the two sentences from the list that state the most important ideas in the passage.

Even though sleep is very important, many people these days are not getting enough.

_____ Some animals sleep up to 18 hours a day.

☒ _____ Modern 24/7 culture is causing more people to suffer from a lack of sleep.

_____ In February of 2009, a commuter plane crashed on its way from Newark, New Jersey, to Buffalo, New York.

_____ Studies show that sleeping for at least eight hours after studying helps students remember what they've learned.

☒ _____ Researchers are learning how sleep deprivation can have a significantly negative effect on performance.

_____ Exercising on a regular basis can help people sleep better, but exercising right before bedtime can make it difficult to fall asleep.

IDENTIFYING REASONS AND EFFECTS

It can be useful to create a summary map of a text, especially to help you understand relationships such as processes, sequences, and reasons and effects. These kinds of relationships frequently occur in scientific articles.

What are some of the reasons for and effects of sleep deprivation?
Complete the summary with information from the reading.

Reasons for sleep deprivation

24/7 expectations

Culture

people want information₁,
entertainment₂, food₃

24 hours a day

Business

employees rewarded for
arriving at work early and staying late₄

Effects of sleep deprivation

Poor performance

School

lower grades₅

Work

Pilots₆ and drivers cause more
fatigue-related accidents/crashes₇, medical
residents make more mistakes₈

UNDERSTANDING STATISTICS

A. Find statistics in the passage to answer each question.

According to the reading, what percentage of ...

1. adults feel sleepy during the day? 20
2. younger adults feel that sleep deprivation affects their work? 50
3. medical residents say that sleepiness led to the death of a patient? 5
4. car accidents occurred because of tired drivers? 20

B. Which of the statistics above do you think are similar in your own country? Which statistic do you find most surprising? Why? Note your ideas and discuss with a partner.

Answers will vary.

UNDERSTANDING INFOGRAPHICS

A. Complete the sentences based on the information in the infographic on page 28.

1. Newborns and infants need more sleep than toddlers.
2. Adults need to sleep the fewest hours.
3. Teenagers need from 8.5 to 9.5 hours of sleep.
4. Four-year-old children need from 11 to 13 hours of sleep.
5. Toddlers need, on average, an hour more sleep than preschoolers.

B. Complete the sentences based on the information in the infographic on page 28.

1. Look at the information for adults' sleep needs. How do you think sleep requirements might vary within this category? For example, do older people need more or less sleep than people in their 30s? Explain.

Answers will vary.

2. What other information might be useful to include in this infographic? How else could the information be organized?

Answers will vary.

BUILDING VOCABULARY

A. Use these words from the passage to complete the paragraph below.

acknowledge consequences crucial injure medical

According to sleep experts, sleep is 1. crucial for our health and performance. When we don't get enough sleep, we are more likely to get sick and make mistakes. Sleep is especially important for people who work in the 2. medical field. Doctors and nurses often work long hours. When they are sleepy, they might accidentally 3. injure a patient or even cause a patient's death. Even though most people know that sleep is important, they continue to deprive themselves of it. This is why it's important to 4. acknowledge that lack of sleep is a significant problem. The 5. consequences of sleep deprivation are just too serious to ignore.

B. Circle the words that are closest in meaning to the words in bold. Some items have more than one answer.

1. **institute**

- a. team b. class **c. organization** d. classroom

2. **investigation**

- a. research** **b. examination** c. lawsuit d. problem

3. **legal**

- a. criminal b. difficult **c. lawful** **d. allowed**

4. **ultimately**

- a. finally** b. quickly **c. eventually** d. realistically

5. **apparently**

- a. factually **b. seemingly** c. certainly d. enthusiastically

C. Have there been any investigations reported in the news recently? Share your ideas with a partner.

Answers will vary.

GETTING MEANING FROM CONTEXT

A. Find these words and phrases (1–7) in the reading on pages 26–29. Match each one with the best definition (a–g).

- | | |
|---|---|
| <u> a </u> 1. totally deprived of shut-eye
(paragraph 2) | a. suffering from complete lack of sleep |
| <u> c </u> 2. skimping (paragraph 2) | b. understand something in a different way |
| <u> e </u> 3. round-the-clock (paragraph 3) | c. not doing or getting as much as you should |
| <u> f </u> 4. intoxicated (paragraph 4) | d. very probably |
| <u> d </u> 5. most likely (paragraph 5) | e. all day; 24 hours a day |
| <u> g </u> 6. fatigue (paragraph 5) | f. affected by alcohol or drugs; drunk |
| <u> b </u> 7. change our perspective
(paragraph 7) | g. a feeling of tiredness |

B. Can you think of a time when you “changed your perspective” on something? What caused it to change? Share your answer with a partner.

Answers will vary.

CRITICAL THINKING

1. **Analyzing.** Use the chart to analyze three of the writer’s claims. Are the writer’s claims supported with quotes, statistics, or information from or about experts?

Claim	Quote?	Stat?	Expert?
Americans are sleeping less than they used to.	✓	✓	✓
Sleep deprivation is partly due to a 24/7 culture.			✓
Sleep deprivation can be deadly.		✓	✓

2. **Evaluating.** What else could be included to make the writer’s claims more convincing? Do you think the writer’s overall argument is convincing? Why or why not?

Answers will vary.

EXPLORE MORE

Read more about why we need sleep at ngm.nationalgeographic.com/2010/05/sleep/max-text. Share what you learn with the class.

A boy yawns as he waits for customers at his roadside apple stall in Kabul, Afghanistan. ►





HOW TO SUCCEED? GET MORE SLEEP

ARIANNA HUFFINGTON Journalist, TED speaker

The lifestyle of a globetrotting media executive doesn't lend itself to getting deep, regular sleep. Arianna Huffington knows about this firsthand.

Huffington is the cofounder and editor-in-chief of *The Huffington Post*, one of the most popular news and blog sites on the Internet. She is also a journalist, the author of several books, and the host of popular political radio shows.

Born in Athens, Greece, Huffington moved to the United Kingdom at the age of 16. She attended the University of Cambridge, where she studied economics. She is now considered by many to be one of the most influential women in media. In 2014, she was included in *Forbes* magazine's list of "The World's 100 Most Powerful Women."



Huffington's career has brought her money and power, the two traditional "metrics of success," but recent experiences with sleep deprivation and exhaustion have caused Huffington to consider a possible third indicator of success. In her book *Thrive: The Third Metric to Redefining Success and Creating a Life of Well-Being, Wisdom, and Wonder*, Huffington explores the idea that true success requires mental, physical, and emotional health and well-being.

influential: *adj.* having the power to affect or control something or someone

lend itself: *idiom* be suitable for (a particular purpose)

Arianna Huffington's **idea worth spreading** is that a good night's rest is a simple but powerful way to increase our productivity and happiness.



In this lesson, you are going to watch Arianna Huffington's TED Talk. Use the information on page 34 to answer each question.

1. Where was Huffington born, and where did she go to college?

Huffington was born in Athens, Greece, and went to the University of Cambridge in England.

2. What field of work is Huffington in now?

media and journalism

3. What is the main message of Huffington's book *Thrive*?

That good health and balance are also keys to success

4. Why do you think she is regarded as one of the world's "most powerful women"?

Her news and blog site is one of the most viewed in the world. People are also exposed to her ideas through her articles, her books, and her radio talk shows.

PART 1

A JOURNEY OF REDISCOVERY

PREVIEWING

A. Read these excerpts from Huffington's talk. Then discuss your answers to the questions below with a partner.

"My big idea is a very, very small idea that can unlock billions of big ideas that are at the moment dormant inside us. And my little idea that will do that is sleep."

"Two and a half years ago, I fainted from exhaustion. I hit my head on my desk. I broke my cheekbone, I got five stitches on my right eye."

dormant: *adj.* hidden, unexpressed, undeveloped

1. Why do you think Huffington says her idea will unlock billions of big ideas? Whose ideas is she referring to?

She means that people who are sleep-deprived are not able to have good ideas. By sleeping more, our creativity will thrive. She is referring to the ideas that are inside everyone, which never emerge because we are too tired to be innovative.

2. Why do you think Huffington tells the story about fainting?

She is explaining why and how she began to fully understand and appreciate the power of sleep.

UNDERSTANDING MAIN IDEAS

A. Read the excerpt from Huffington's talk. Choose the sentence that best paraphrases the excerpt.

"I began the journey of rediscovering the value of sleep. And in the course of that, I studied, I met with medical doctors, scientists, and I'm here to tell you that the way to a more productive, more inspired, more joyful life is getting enough sleep.."

- a. I did a lot of research on sleep and discovered that many medical doctors and scientists suffer from a serious lack of sleep.
- b. I participated in sleep experiments with doctors and scientists, and I discovered that I had problems sleeping.
- ☒ c. I studied a lot about the importance of sleep, and I discovered that getting enough sleep can lead to a better life.

B. Write an answer to the question below.

According to Huffington, what are some benefits of getting enough sleep?

With more sleep, people become more productive, more inspired, and more joyful.

UNDERSTANDING A CLASSIFICATION

Arianna Huffington refers to Type A people in her talk. Read the description of personality types, and discuss answers to the questions with a partner.

The Type A/Type B personality theory was developed by cardiologists Meyer Friedman and Ray Rosenman in the 1950s. The two doctors developed the theory after observing patients with heart conditions in the waiting room of their office. They noticed that some of the heart patients were unable to sit still, and these patients wore out the seats of the chairs by moving around nervously. Friedman and Rosenman labeled these people Type A and determined that a Type A person is more likely than another person to have heart disease and high blood pressure.

According to the theory, Type A people tend to be very competitive and rushed and they become angry more easily than other people. These individuals often have high-stress jobs and spend a lot of time working and striving to succeed. Type B individuals tend to be more relaxed, less anxious, and more creative and imaginative. There is also a Type C personality. These individuals tend to hold their emotions in, particularly negative ones like anger and frustration. They also avoid conflict and try to be nice to everyone.

Friedman and Rosenman's theory is not perfect—not everyone falls into one of these categories. However, people often use the term “Type A” to describe high-achieving, competitive individuals who lead stressful lives.

1. Which type of person do you think is the most likely to sleep well?

Type B because they worry the least

2. Which type do you think you are? Explain.

Answers will vary.

3. Think of people you know who fit into these three categories. Describe them.

Answers will vary.

4. Huffington says that most of her audience are “Type A” people. Why does she think they are especially sleep-deprived?

Type A people are the most likely to value work over sleep, and therefore not get enough sleep.

PART 2

A NEW REVOLUTION

UNDERSTANDING METAPHORS

- A.** Huffington uses a metaphor (comparison) to make a point about leadership. Read the excerpt and complete the sentence to explain what she means.

“ . . . the essence of leadership is being able to see the iceberg before it hits the *Titanic*. And we’ve had far too many icebergs hitting our *Titanics*.”

What Huffington means is that a good leader should be able to _____.

- a. see a problem before it happens
- b. come up with creative solutions after a problem occurs

B. Watch (▶) the next segment of Huffington's TED Talk to check your ideas.



UNDERSTANDING A SPEAKER'S MESSAGE

A. Huffington believes that men and women value sleep differently.
Answer the questions and share your ideas with a partner.

1. Huffington says, "Sleep deprivation has become a virility symbol." What does she mean by *a virility symbol*? How might sleep deprivation be a virility symbol for some men?

A virility symbol is

something that demonstrates to others that you are powerful and important.

Some men try to express their virility by saying

they don't sleep a lot.

2. Huffington tells a story about a dinner date with a man who bragged about getting only four hours of sleep the night before. Why do you think she tells this story?

She says that the date wasn't very interesting, by which she is trying to illustrate that the man would have been a better date if he had slept more.

B. In the excerpt below, Huffington refers to Lehman Brothers (a large American financial services company that declared bankruptcy in 2008). What point is she making?

“In fact, I have a feeling that if Lehman Brothers was Lehman Brothers and Sisters, they might still be around. While all the brothers were busy just being hyper-connected 24/7, maybe a sister would have noticed the iceberg, because she would have woken up from a seven-and-a-half or eight-hour sleep and have been able to see the big picture.”

Her point is that most women are more likely to recognize the power of sleep, and perhaps some major world crises could have been avoided if more well-rested women were in places of power, instead of sleep-deprived men.

CRITICAL THINKING

1. **Evaluating.** Do you agree or disagree with the way Huffington characterizes men and women in regards to sleep? Why or why not?

Answers will vary.

2. **Synthesizing.** How does Arianna Huffington's TED Talk support the main ideas of the reading passage on pages 26–29?

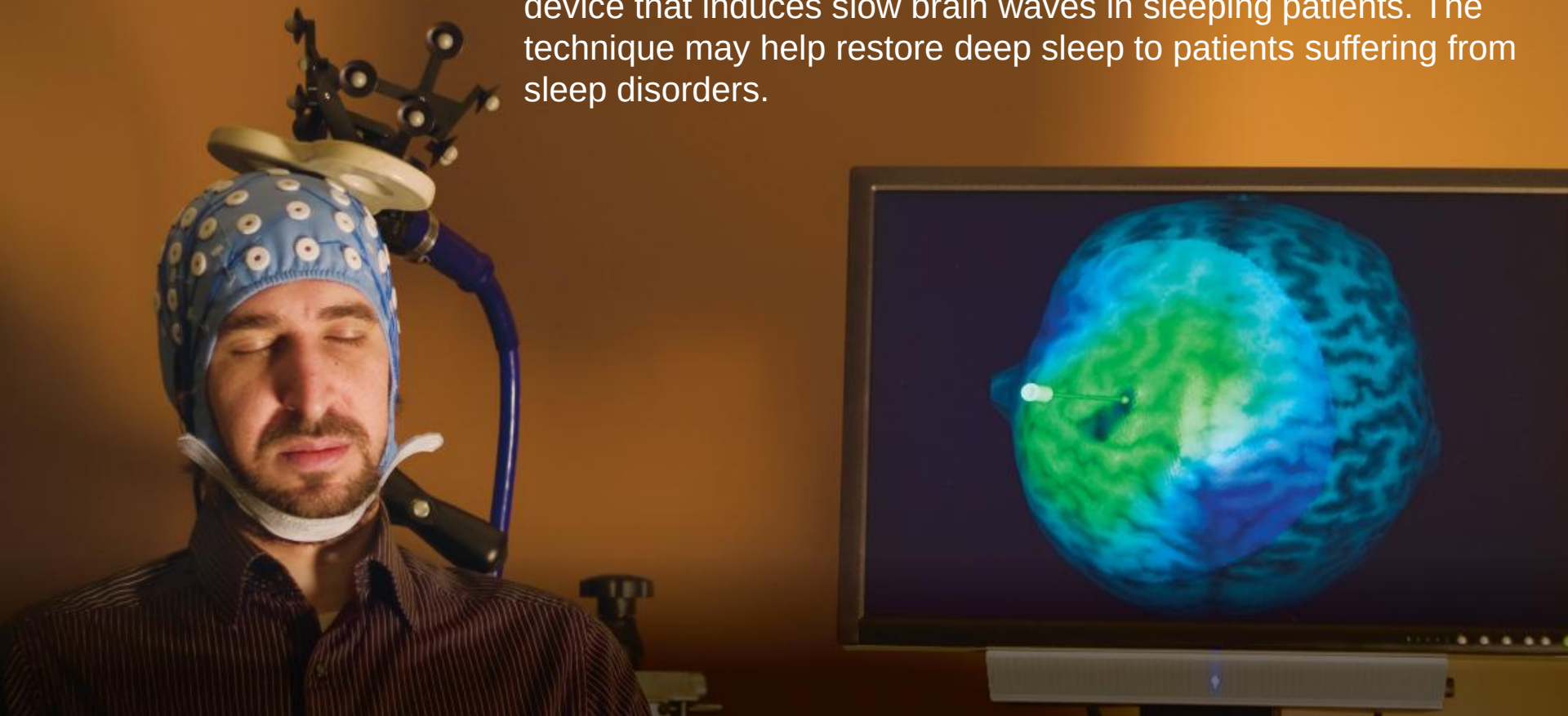
She offers the same message: Sleep deprivation is hurting global society and is dangerous. She additionally points out the positive side of getting more sleep, most notably how it can actually make us more productive and help us enjoy life more.

EXPLORE MORE

Learn about how our internal body clock can regulate our sleep schedule by watching Jessa Gamble's TED Talk "Our natural sleep cycle" on TED.com. Share what you learn with the class.

Project

A researcher at the University of Wisconsin demonstrates a device that induces slow brain waves in sleeping patients. The technique may help restore deep sleep to patients suffering from sleep disorders.



Project

A. Work with a partner. You are going to find out more about one of the sleep-related topics below.

- The effects of modern technology (e.g., tablets, smartphones, television, computers) on sleep
- Ways to improve sleep
- Sleepwalking
- Other sleep disorders
- Why we dream
- (Your own idea):

_____.

1. With your partner, choose the topic that you find the most interesting or come up with your own sleep-related topic.
2. Go online to learn as much as you can about the topic. Take notes on the most important and interesting information.
3. Use your information to create a two-minute presentation. You can use drawings, photos, and video to explain your information. Where possible, include real-life examples, case studies, and anecdotes to illustrate your ideas.

Project

B. Work with two other pairs.

- Give your presentation to the other pairs.
- As you listen to the other pairs' presentations, take note of the key points. Review the information at the end of the presentations.

C. Discuss these questions with your class.

1. What new information did you learn? What was most surprising?
2. What topic do you want to learn more about?

EXPLORE MORE

Learn more about sleep by watching Russell Foster's TED Talk "Why do we sleep?" at TED.com. How does Foster describe the purpose of sleep? Share your information with the class.

Part 1

My big idea is a very, very small idea that can unlock billions of big ideas that are at the moment dormant inside us. And my little idea that will do that is sleep.

This is a room of type-A women. This is a room of sleep-deprived women. And I learned the hard way, the value of sleep. Two-and-a-half years ago, I fainted from exhaustion. I hit my head on my desk. I broke my cheekbone. I got five stitches on my right eye. And I began the journey of rediscovering the value of sleep. And in the course of that, I studied, I met with medical doctors, scientists, and I'm here to tell you that the way to a more productive, more inspired, more joyful life is getting enough sleep.

Part 2

And we women are going to lead the way in this new revolution, this new feminist issue. We are literally going to sleep our way to the top, literally.

Because unfortunately for men, sleep deprivation has become a virility symbol. I was recently having dinner with a guy who bragged that he had only gotten four hours' sleep the night before. And I felt like saying to him—but I didn't say it—I felt like saying, "You know what? If you had gotten five, this dinner would have been a lot more interesting."

There is now a kind of sleep deprivation one-upmanship. Especially here in Washington, if you try to make a breakfast date, and you say, "How about eight o'clock?" They're likely to tell you, "Eight o'clock is too late for me, but that's OK, you know, I can get a game of tennis in and do a few conference calls and meet you at eight." And they think that means that they are so incredibly busy and productive, but the truth is they're not, because we, at the moment, have had brilliant leaders in business, in finance, in politics, making terrible decisions. So a high I.Q. does not mean that you're a good leader, because the essence of leadership is being able to see the iceberg before it hits the Titanic. And we've had far too many icebergs hitting our *Titanics*.

In fact, I have a feeling that if Lehman Brothers was Lehman Brothers and Sisters, they might still be around. While all the brothers were busy just being hyper-connected 24/7, maybe a sister would have noticed the iceberg, because she would have woken up from a seven-and-a-half- or eight-hour sleep and have been able to see the big picture.

So as we are facing all the multiple crises in our world at the moment, what is good for us on a personal level, what's going to bring more joy, gratitude, effectiveness in our lives and be the best for our own careers is also what is best for the world. So I urge you to shut your eyes and discover the great ideas that lie inside us, to shut your engines and discover the power of sleep.

Thank you.